2026 MIDDLE SCHOOL STEM CAMPS

PY-STEM

Instructor Guide Packet

Facilitation guides for every primary and backup activity.

Magnetic Capsule Maze Cup

Steer a pill-camera without touching it

At a glance: Steer a magnetic pill-camera through a body maze without touching it. About 80 min; 4 to 6 teams.

MATERIALS

- small neodymium magnets
- maze boards
- steel tokens or paper clips
- timers
- barrier sheets

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does moving the magnet slowly help?
- How does distance change the pull you feel?
- Where is remote actuation used in real medicine?

SCORING (100 POINTS)

Scoring criterion	Points
Fastest clean run	50
Low penalties	20
Magnetic-control explanation	20
Redesign	10
Total	**100**

Sources: Science Museum Group Magnetic Maze

Oobleck Armor Arena

Armor from a liquid that hardens on impact

At a glance: Design armor with a material that acts like a liquid until impact. About 80 min; 4 to 6 teams.

MATERIALS

- cornstarch
- water
- zip bags
- cardboard sleeves
- test weights
- spoons
- trays

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DEBRIEF

- Why does a fast press feel harder than a slow one?
- How little oobleck still protected the target?
- Where would shear-thickening armor be useful?

SCORING (100 POINTS)

Scoring criterion	Points
Protection result	40
Material efficiency	20
Explanation	20
Redesign	10
Cleanup	10
Total	**100**

Sources: Science Buddies Oobleck; Exploratorium Ooze

Cardboard Automata Arcade

One crank, one clever motion

At a glance: Build a one-crank machine that performs a mini task. About 80 min; 4 to 6 teams.

MATERIALS

- cardboard
- skewers
- straws
- craft sticks
- tape
- hot glue station
- scissors

FACILITATION ARC

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DEBRIEF

- How did your cam shape change the motion?
- What caused jams, and how did you fix them?
- Where do cams show up in real machines?

SCORING (100 POINTS)

Scoring criterion	Points
Reliable motion	30
Cam or linkage complexity	20
Task success	20
Explanation	20
Aesthetics and build quality	10
Total	**100**

Sources: Exploratorium Cardboard Automata

INSTRUCTOR · PYS-04 · BIOMEDICAL

Stethoscope Sprint and Recovery

Build a listening tool, measure like a scientist

At a glance: Build a medical listening tool and use it like a sports scientist. About 80 min; 4 to 6 teams.

MATERIALS

- funnels
- flexible tubing
- tape
- balloons or membranes
- stopwatches
- sanitizing wipes

FACILITATION ARC

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DEBRIEF

- What change most improved your sound quality?
- Why does a good seal matter so much?
- What does a fast recovery suggest?

SCORING (100 POINTS)

Scoring criterion	Points
Sound quality	30
Measurement accuracy	25
Explanation	20
Redesign	15
Data clarity	10
Total	**100**

Sources: Science Buddies Make a Stethoscope

INSTRUCTOR · PYS-05 · BIOMEDICAL

Reaction Time Combine

Prove which strategy cuts your delay

At a glance: Train like a sports scientist and prove which strategy cuts delay. About 80 min; 4 to 6 teams.

MATERIALS

- meter sticks
- stopwatches
- cones
- score sheets
- clipboards

FACILITATION ARC

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DEBRIEF

- Why use the median instead of your fastest catch?
- Which strategy helped, and why might it?
- Where does reaction time matter in real life?

SCORING (100 POINTS)

Scoring criterion	Points
Best median reaction time	35
Data quality	20
Strategy improvement	20
Explanation	15
Teamwork	10
Total	**100**

Sources: Science Buddies Reaction Time

SONAR Slinky Showdown

Make invisible distance sensing visible

At a glance: Make invisible distance sensing visible with a slinky pulse. About 80 min; 4 to 6 teams.

MATERIALS

- 4 to 6 metal slinkies
- floor tape
- station cards
- clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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DEBRIEF

- How is a slinky pulse like a sound wave?
- Why does timing an echo give distance?
- Where are ultrasonic sensors used?

SCORING (100 POINTS)

Scoring criterion	Points
Model and predictions	35
Station performance	25
Explanation	20
Team communication	20
Total	**100**

Sources: IIHS SONAR Slinky

INSTRUCTOR · PYS-07 · OPTICS + LIGHT

Pinhole Precision Challenge

The sharpest image, no lens

At a glance: Build the sharpest camera image with no lens. About 80 min; 4 to 6 teams.

MATERIALS

- cardstock
- foil
- tape
- pushpins or paper clips
- white screen paper
- light source

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why is the image upside down?
- What did shrinking the hole do, and why?
- How is this related to your eye?

SCORING (100 POINTS)

Scoring criterion	Points
Image clarity	35
Concept explanation	25
Build quality	20
Redesign	10
Speed	10
Total	**100**

Sources: Exploratorium Personal Pinhole Theater; NASA JPL Pinhole Camera

Low-Ropes Force Map Relay

Predict balance before the element proves you right

At a glance: Predict balance and stability before the ropes element proves you right. About 80 min; 4 to 6 teams.

MATERIALS

- element cards
- clipboards
- center-of-mass templates
- pencils

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
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DEBRIEF

- When did your prediction match what you felt?
- How did lowering your center of mass help?
- Where else does balance physics matter?

SCORING (100 POINTS)

Scoring criterion	Points
Correct predictions	35
Force-map quality	25
Debrief explanation	20
Teamwork reflection	20
Total	**100**

Sources: Temple Ambler Outdoor Experiential Education; TeachEngineering balance references

INSTRUCTOR · PYS-09 · MECHANICS

Hovercraft Hockey Hackathon

A puck that barely touches the floor

At a glance: Build a puck that moves by barely touching the floor. About 80 min; 4 to 6 teams.

MATERIALS

- recycled CDs or plastic discs
- pop-top bottle caps
- balloons
- hot glue station
- targets
- floor tape

FACILITATION ARC

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DEBRIEF

- Why does the air cushion reduce friction?
- Why is more lift not always better?
- Where are air bearings used in real life?

SCORING (100 POINTS)

Scoring criterion	Points
Glide performance	30
Target competition score	25
Explanation	20
Redesign	15
Build quality	10
Total	**100**

Sources: NASA JPL Hovering on a Cushion of Air; Science Buddies Hovercraft

Spectra Sleuth Showdown

Two white lights, two hidden color fingerprints

At a glance: Can two white lights hide different color fingerprints? About 80 min; 4 to 6 teams.

MATERIALS

- diffraction glasses or gratings
- LED sources
- incandescent source if available
- spectrum cards
- museum clue sheets

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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DEBRIEF

- How can two white lights have different spectra?
- What does a line spectrum tell you?
- How do chemists make firework colors?

SCORING (100 POINTS)

Scoring criterion	Points
Correct matches	40
Spectrum sketches	20
Explanation	20
Exhibit connection	20
Total	**100**

Sources: Science History Institute Flash Bang Boom; Exploratorium Spectra

BookBot Bin Logic Challenge

Retrieve by address, not by conveyor race

At a glance: Retrieve the right book by using bin addresses, not a conveyor race. About 80 min; 4 to 6 teams.

MATERIALS

- bin cards
- address labels
- order deck
- route mat
- timers
- optional grabber tool

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
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DEBRIEF

- Why store by address instead of by subject?
- What made one route faster than another?
- Where else is this storage logic used?

SCORING (100 POINTS)

Scoring criterion	Points
Correct retrievals	35
Low traffic conflicts	25
Routing or algorithm explanation	20
Efficiency	10
Teamwork	10
Total	**100**

Sources: Temple Charles Library BookBot; Charles Library Circulating Collections

INSTRUCTOR · PYS-12 · DESIGN

Accessibility Ramp Rescue Lab

A portable ramp that meets real constraints

At a glance: Design a portable ramp that works for a real user constraint. About 80 min; 4 to 6 teams.

MATERIALS

- foam board
- craft sticks
- dowels
- binder clips
- rulers
- weighted cart
- tape

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
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DEBRIEF

- Why does a gentler slope need more length?
- Which criterion was hardest to meet?
- Why design for accessibility from the start?

SCORING (100 POINTS)

Scoring criterion	Points
Load held	30
Slope meets target	30
Portability	20
Client criteria explanation	20
Total	**100**

Sources: TeachEngineering Wheelchair Ramp Design

BACKUP ACTIVITIES

Ready alternates

Pencil Pressure Safety Lab

Spread a force before it dents the target

At a glance: Spread a force before it dents the target. About 60 min; 4 to 6 teams.

MATERIALS

- Pencils
- Foam
- Cardboard
- Weights
- Graph paper

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does the same force do less damage over more area?
- Where is this used in safety gear?
- What made your spreader better?

SCORING (100 POINTS)

Scoring criterion	Points
Pressure spread	40
Explanation	25
Data quality	20
Redesign	15
Total	**100**

Sources: IIHS Pencil Pressure

INSTRUCTOR · PYB-02 · BIOMEDICAL

Domino Neuron Relay

Make a nerve signal survive a gap

At a glance: Make a nerve signal survive a gap. About 60 min; 4 to 6 teams.

MATERIALS

- Dominoes
- Rulers
- Tape
- Index cards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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- Redesign one variable, re-test, compare to baseline.
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DEBRIEF

- What happened when the gap got too wide?
- How is a bridge piece like a synapse?
- Why does myelin speed real signals?

SCORING (100 POINTS)

Scoring criterion	Points
Transmission success	35
Gap challenge	25
Explanation	20
Redesign	20
Total	**100**

Sources: Exploratorium Domino Effect

Pulley Rescue Relay

Lift the load with smarter force, not stronger arms

At a glance: Lift the load with smarter force, not stronger arms. About 60 min; 4 to 6 teams.

MATERIALS

- Classroom pulleys
- String
- Weights
- Spring scales

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
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DEBRIEF

- Why did the movable pulley reduce effort?
- What did you trade for the easier lift?
- Where are pulley systems used in rescue work?

SCORING (100 POINTS)

Scoring criterion	Points
Performance	35
Setup accuracy	25
Explanation	20
Teamwork	20
Total	**100**

Sources: TryEngineering Pulleys and Force

Barcode Checksum Rescue

Catch the fake barcode before the wrong part ships

At a glance: Catch the fake barcode before the warehouse ships the wrong part. About 60 min; 4 to 6 teams.

MATERIALS

- Printed barcode cards
- Dry erase boards
- Timers

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
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DEBRIEF

- How does one extra digit catch a typo?
- Why do false alarms matter in a warehouse?
- Where do you see check digits in daily life?

SCORING (100 POINTS)

Scoring criterion	Points
Correct detections	45
Low false alarms	20
Rule explanation	20
Teamwork	15
Total	**100**

Sources: Computer science unplugged error-detection concepts